

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 38)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01 . 1

Distinguish between **renewable** and **depletable** energy resources, giving an example of a resource that is both renewable and depletable.

[3 marks]

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01 . 2

Explain why the abundance of an energy resource is not the same as the amount available for use. Give **two** examples.

[3 marks]

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01 . 3

Describe the locational constraints of **two** named renewable energy resources.

[4 marks]

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10

- 0 2** Figure 1 lists the stages in a water treatment works supplying drinking water from a lowland river.

Figure 1

Stage	Process
1	Screens
2	Sedimentation
3	Aeration
4	Flocculation and clarification
5	Filtration through sand and gravel
6	Activated carbon
7	Sterilisation
8	pH adjustment

- 0 2** . **1** Explain why clay particles do not settle out in the sedimentation stage, and describe how they are removed in stage 4.

[3 marks]

- 0 2** . **2** Explain the purpose of aeration (stage 3).

[2 marks]

- 0 2** . **3** Explain why ozone or UV light may be used for sterilisation instead of chlorine when water comes from peaty upland sources.

[2 marks]

0 2 . 4 Describe how each of the following contaminants is removed from water. **[3 marks]**

Toxic metal ions (lead, mercury, arsenic)

Salt

Odours and pesticides

10

0 3 A motor-racing circuit has residents living 500 m away. The circuit operator wants to compare two designs of acoustic barrier: an earth baffle mound and a timber acoustic fence.

0 3 . 1 Describe how you would carry out an investigation to compare the effectiveness of the two barriers. **[5 marks]**

03 . 2

Explain why readings should be taken with the same wind direction.

[1 mark]

03 . 3

Explain the difference between the LA10 and LA90 measures of traffic noise.

[2 marks]

03 . 4

Outline **two** ways in which the residents' houses could be designed to reduce the noise reaching occupants.

[2 marks]

1

2

10

04

A student compared the invertebrates in a stream above and below a sewage outfall using kick sampling.

04 . 1

Describe how kick sampling is carried out so that the samples from the two sites are comparable.

[3 marks]

0 4 . 2

Give **three** limitations of kick sampling.

[3 marks]

1

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2

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3

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0 4 . 3

Explain how a Surber sampler improves on kick sampling.

[2 marks]

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0 4 . 4

Explain why the invertebrates found give a better indication of long-term sewage pollution than a single measurement of dissolved oxygen.

[2 marks]

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0 5

A nuclear reprocessing plant on the coast discharges low-level liquid radioactive waste into the sea through a pipeline.

0 5 . 1

Explain what is meant by **Critical Pathway Analysis** (CPA).

[2 marks]

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0 5 . 2

Give **three** factors that would be considered in a CPA for this discharge.

[3 marks]

1

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2

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3

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0 5 . 3

Describe how **Critical Group Monitoring** (CGM) would be used to assess the risk to the public from this discharge.

[3 marks]

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0 5 . 4

Suggest **two** types of environmental sample that would be collected to check the CPA predictions.

[2 marks]

1

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2

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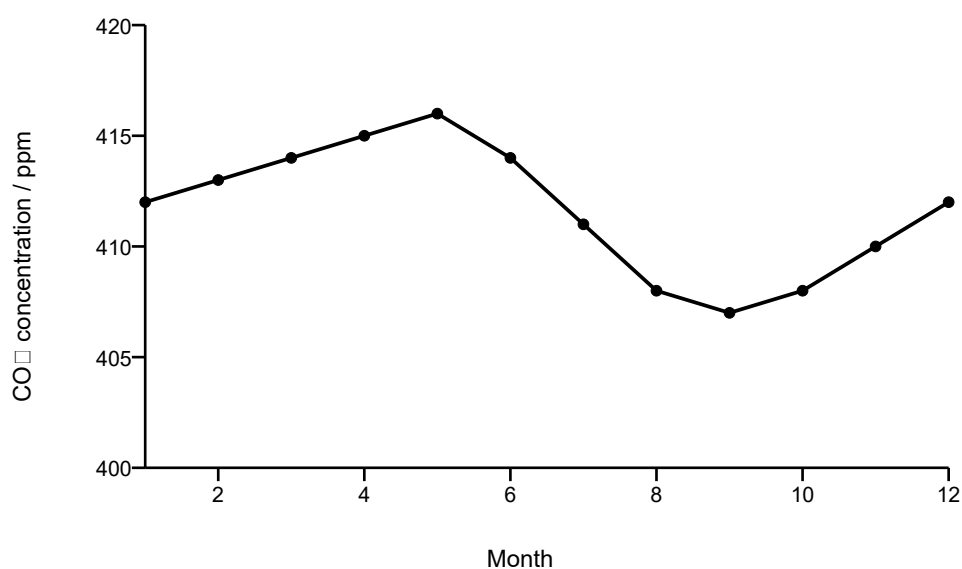
10

0 6

Table 1 shows the composition of the dry atmosphere. Figure 1 shows the changes in carbon dioxide concentration measured at a site in the northern hemisphere over one year.

Table 1

Gas	Percentage by volume / %
Nitrogen	78
Oxygen	21
Argon and other rare gases	0.9
Carbon dioxide	0.04
Ozone	0.000007

Figure 1**0 6****. 1**

Describe and explain the pattern shown in Figure 1.

[4 marks]

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06 . 2

Explain why the atmosphere is described as being in a state of **dynamic equilibrium**.

[2 marks]

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06 . 3

Give **two** ways in which the atmosphere supports life other than by providing gases for biological molecules.

[2 marks]

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2

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06 . 4

Name the two layers of the atmosphere most affected by human activities.

[2 marks]

1

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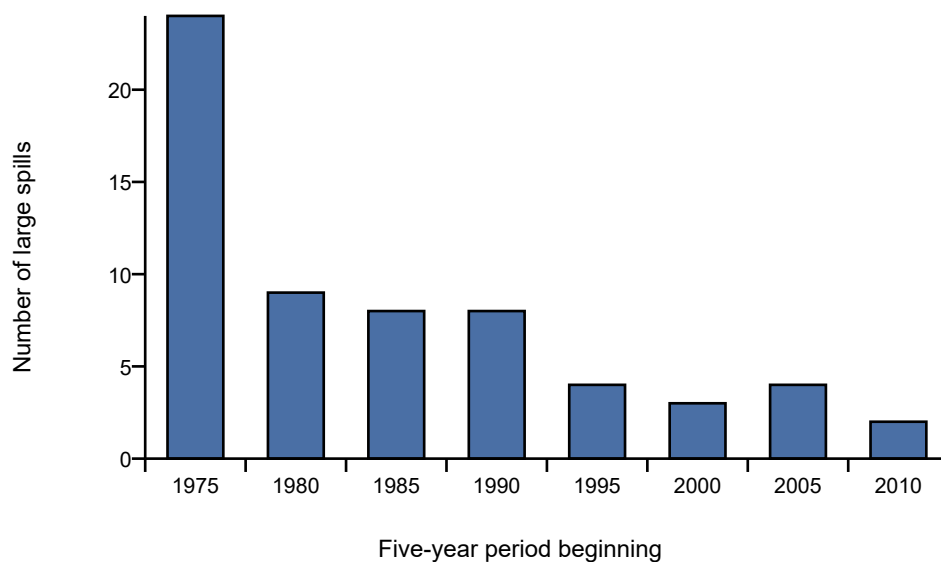
2

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10

07

Figure 1 shows the number of large oil spills (over 700 tonnes) from tankers worldwide and the amount of oil transported by sea between 1975 and 2015.

Figure 1**Table 1**

Year	1975	1985	1995	2005	2015
Oil transported by sea / million tonnes	1 400	1 200	1 800	2 400	2 900

07**1**

Calculate the percentage increase in oil transported by sea between 1975 and 2015.

[2 marks]**Answer**

%**07****2**

Describe the trend in the number of large oil spills shown in Figure 1.

[2 marks]

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07 . 3

Suggest **three** reasons for the decline in the number of large spills despite the increase in oil transported.

[3 marks]

1

2

3

07 . 4

Explain why the total amount of oil pollution entering the sea may not have fallen as much as the number of large tanker spills.

[3 marks]

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10

08 . 1

Define the term **persistence** as applied to a pollutant.

[1 mark]

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08 . 2

Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

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5

09 . 1

Describe **three** factors that have restricted the growth of nuclear power.

[3 marks]

1

2

3

09 . 2

Explain how a plutonium 'breeder' reactor can produce more fissile fuel than it uses.

[3 marks]

09 . 3

Give **two** advantages and **two** disadvantages of thorium reactors compared with uranium reactors.

[4 marks]

Advantages

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Disadvantages

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10

1 0 . 1

Evaluate the success of the methods and strategies used to reduce global climate change.

[9 marks]

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1 0 . 2

[1 mark]

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10

Write an essay on one of the following topics.

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Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

OR

1 1 . 2

Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

This image shows a full page of primary-ruled paper. It features a series of horizontal dashed lines spaced evenly down the page, designed for handwriting practice. A single solid line runs horizontally across the top of the page, serving as a baseline or margin. The rest of the page is white and contains no other markings or text.

END OF QUESTIONS